

AT-STLN-ARTIX 7-001

FPGA Development Board

Hardware Reference Manual

Anmaya Technologies

Product Code: ANM-PRD-2025-005 | Rev: 1.0

Schematic Tool: KiCad E.D.A. 9.0.4

1. Board Overview

The AT-STLN-ARTIX 7-001 is a feature-rich FPGA development board designed by Anmaya Technologies for educational and prototyping purposes. It is built around the Xilinx XC7A35T-FTG256-1 (Artix-7 series) FPGA and integrates a wide range of peripherals, actuators, and communication interfaces on a single PCB.

1.1 Key Features

- FPGA: Xilinx XC7A35T-FTG256-1 (Artix-7, 33K logic cells, 256-ball FTBGA, Speed Grade -1)
- USB Interface: FTDI FT232H high-speed USB-to-JTAG/UART bridge
- Configuration: MX25L3233FM1 32Mb SPI Flash for bitstream storage
- Memory: IS61WV25616EDBL 512K x 16-bit asynchronous SRAM
- DAC: AD9744 14-bit, 210 MSPS high-speed DAC with OPA356AID output op-amp
- ADC: MCP3202 12-bit, 2-channel SPI ADC
- Display: 16x2 character LCD (DS1WC1602A) + 4-digit 7-segment display (MAX7219 driver)
- Wireless: ESP32-C3-MINI-1 Wi-Fi/BLE co-processor
- Motors: DC motor driver (DRV8833) + stepper motor driver (L293DD)
- I/O: PMOD connector (20-pin, 16 user I/Os), 4x4 matrix keypad, 16 user LEDs, 8 slide switches, 6 push buttons
- Relay: Single SANYOU SRD Form-C relay (5V coil, 10A contacts)
- Buzzer: Active piezo buzzer (BC817 NPN driver)
- Storage: Micro SD card slot
- Power: Multi-rail power supply (5V, 3.3V, 1.8V, 1.0V) from 15V DC barrel jack
- Clock: 24 MHz system oscillator (ASDM1) + 12 MHz USB crystal

2. FPGA Specifications

2.1 XC7A35T-FTG256-1 Key Parameters

Parameter	Value
Device Family	Xilinx Artix-7
Part Number	XC7A35T-FTG256-1
Speed Grade	-1 (slowest / most power-efficient commercial grade)
Package	FTG256 (FTBGA, 256-ball, 17×17 mm)
Logic Cells	33,280
CLB Flip-Flops	41,600 (20,800 LUTs)
CLB Slices	8,150
Distributed RAM	400 Kb

Block RAM (36Kb tiles)	50 tiles (1,800 Kb total)
DSP48E1 Slices	90
XADC (12-bit ADC)	1 block (dual 12-bit, 1 MSPS, up to 17 channels)
Max User I/O	170
PLLs / MMCMs	2 PLLs, 2 MMCMs
VCCINT (Core)	1.0 V
VCCAUX	1.8 V
VCCO (I/O Banks)	3.3 V (banks 14, 15, 34, 35)
Configuration	Master SPI (M[2:0]=001) / JTAG (M[2:0]=101)
SPI Flash	MX25L3233FM1 (32Mb, 3.3V)
Temperature Range	0°C to +85°C (Commercial grade)

3. FPGA Pin Mapping

The following tables list the complete FPGA pin assignments derived from the schematic (ANM-PRD-2025-005). The FPGA has four active I/O banks used for user logic: Banks 14, 15, 34, and 35.

3.1 User LEDs (Generic)

Net Name	FPGA Pin (Bank 35)
LED1	D5
LED2	A3
LED3	B4
LED4	A4
LED5	E6
LED6	C13
LED7	C14
LED8	D14

3.2 7-Segment Display (MAX7219 — SPI Interface)

The 4-digit 7-segment display (U15, XLITX 3641AS) is driven by a MAX7219 (U14) via SPI. The FPGA drives 3.3V signals through SN74LVC1T45 level-shifters (U17/U18/U19) to the 5V MAX7219.

Signal	FPGA Net	FPGA Pin (Bank 15)	Direction	Description
7SEG_{DIN}	7SEG_DIN	J15	FPGA → MAX7219	Serial data in
7SEG_{LOAD}	7SEG_LOAD	J16	FPGA → MAX7219	Chip select / load
7SEG_{CLK}	7SEG_CLK	H12	FPGA → MAX7219	Serial clock

4.3 16x2 LCD Interface

The LCD (DS1WC1602A) operates on 5V. It is connected in 8-bit mode to FPGA Bank 35 I/Os. Contrast is set by potentiometer RV1.

LCD Pin	Net Name	FPGA Pin (Bank 35)	Direction
RS (Pin 4)	LCD_RS	G4	FPGA → LCD
R/W (Pin 5)	LCD_R/W	H3	FPGA → LCD
E (Pin 6)	LCD_E	E1	FPGA → LCD
D0 (Pin 7)	LCD_D0	G2	FPGA ↔ LCD
D1 (Pin 8)	LCD_D1	G1	FPGA ↔ LCD
D2 (Pin 9)	LCD_D2	H5	FPGA ↔ LCD
D3 (Pin 10)	LCD_D3	H4	FPGA ↔ LCD
D4 (Pin 11)	LCD_D4	J5	FPGA ↔ LCD
D5 (Pin 12)	LCD_D5	J4	FPGA ↔ LCD
D6 (Pin 13)	LCD_D6	H2	FPGA ↔ LCD
D7 (Pin 14)	LCD_D7	H1	FPGA ↔ LCD

3.4 Slide Switches (SW0–SW15)

16 FPGA-connected slide switches (SPDT) with 220Ω series resistors and pull-up/pull-down to 3.3V. Mapped to FPGA Bank 35.

Switch Net	FPGA Pin
SS0	C9
SS1	B9
SS2	G5
SS3	A7
SS4	C7
SS5	A10
SS6	B7
SS7	A8

3.5 4×4 Matrix Keypad

The 4x4 keypad (keys 0-9, A-F) connects to FPGA Bank 35 via 10K pull-up resistors and 100nF debounce caps. Keys are arranged as rows (SW1–SW4) × columns (SW5–SW8 pull groups).

Keypad Net	FPGA Pin	Keypad Net	FPGA Pin
0	A13	8	A5
1	F5	9	C6
2	E3	A	D4
3	F2	B	F4
4	A12	C	B6
5	D6	D	B5
6	D3	E	C4
7	F3	F	E5

3.7 PMOD Connector (J16 — 20 Pin)

The PMOD connector (J16) exposes 16 FPGA I/O signals from Bank 34 with 0Ω series resistors (R112–R127) for easy daughterboard attachment. GND on pin 19, 3.3V on pin 20.

J16 Pin	FPGA Pin	J16 Pin	FPGA Pin
IO_0	T2	IO_8	N1
IO_1	R3	IO_9	N2
IO_2	T3	IO_10	R2
IO_3	T4	IO_11	N3
IO_4	M1	IO_12	P3
IO_5	M2	IO_13	N4
IO_6	P1	IO_14	P4
IO_7	_R1	IO_15	_P5

3.8 SPI Bus (Shared — SPIm)

A shared SPI master bus connects the FPGA to the ADC (MCP3202), SD card, and ESP32. Chip selects are individual.

Signal	FPGA Pin (Bank 15)	Connected To
SPIm_SCK	G11 (IO_25_15)	ESP GPIO5/ADC2_CH0
SPIm_MOSI	G12 (IO_L19N_T3_A21_VREF_15)	MCP3202 DIN, SD CMD, ESP GPIO7
SPIm_MISO	G14 (IO_L19N_T3_A21_VREF_15)	MCP3202 DOUT, SD DAT0, ESP GPIO4
SD_CS	H14 (IO_L24P_T3_RS1_15)	SD card DAT3/CD
ESP_CS	F14 (IO_L21N_T3_DQS_A18_15)	ESP32 GPIO6
ADC uses SD_CS net	See ADC section	MCP3202 ~CS~

3.9 Clock Inputs

Clock Signal	FPGA Pin	Bank	Source	Frequency
GCLK_24MHz	D13 (IO_L12P_T1_MRCC_15)	15	ASDM1-24MHz oscillator (U4)	24 MHz
EMCCLK	M15 (IO_L3N_T0_DQS_EMCCCLK_14)	14	FT232H ACBUS pin	Ext. clock

3.10 FPGA Configuration Pins

Signal	FPGA Ball	Description
CCLK_0	E8	Configuration clock (SPI)
M0_0	M9	Mode select 0 — SW17 (default: 1 = SPI master)
M1_0	M10	Mode select 1 — SW36 (default: 0)
M2_0	M11	Mode select 2 — SW37 (default: 0)
DONE_0	H10	Configuration done indicator (LED D2)
INIT_B_0	K10	Initialization busy (LED D1, active low)
PROGRAM_B_0	L9	Reprogram trigger (BTN2 PROGRAM, active low)
CFGBVS_0	E7	Config bank voltage select — pulled to 3.3V
TDI_0	N7	JTAG Test Data In
TDO_0	N8	JTAG Test Data Out
TMS_0	M7	JTAG Test Mode Select
TCK_0	L7	JTAG Test Clock

4. Peripheral Interfaces

4.1 USB JTAG / UART (FT232H — U2)

The FT232H (U2) provides USB 2.0 HS to JTAG and UART conversion via a USB-C connector (J1). It is configured using a 93LC56B EEPROM (U10). JTAG signals pass through 100Ω series resistors and ESD protection diodes.

FT232H Signal	Net	FPGA Ball	Notes
ADBUS0	JTAG_R_TDI JTAG_TDI →	N7	TCK via 100Ω R130
ADBUS1	JTAG_R_TDO JTAG_TDO →	N8	100Ω R131
ADBUS2	JTAG_R_TMS JTAG_TMS →	M7	100Ω R132
ADBUS3	JTAG_R_TCK JTAG_TCK →	L7	100Ω R133
ACBUS0-9	GPIO	—	Unused / test points
DM/DP	USB_D- / USB_D+	J1 connector	USB 2.0 data pair

4.2 JTAG Header (J3 — 14-Pin)

J3 Pin	Signal	J3 Pin	Signal
1	JTAG_TMS	2	+3V3
3	GND	4	JTAG_TCK
5	GND	6	JTAG_TDO
7	GND	8	JTAG_TDI
9	GND	10	—
11	GND	12	—
13	GND	14	—

4.3 DAC (AD9744 — U24)

The AD9744 is a 14-bit, 210 MSPS current-output DAC. Its digital inputs connect via 22Ω damping resistors to FPGA Bank 14/15. The differential current outputs (IOUTA/IOUTB) feed an OPA356AID op-amp for single-ended voltage output on J5.

DAC Signal	FPGA Net	FPGA Pin	DAC Signal	FPGA Net	FPGA Pin
DAC_D0	R_DAC_D0	H16	DAC_D7	R_DAC_D7	D16
DAC_D1	R_DAC_D1	G15	DAC_D8	R_DAC_D8	B14
DAC_D2	R_DAC_D2	G16	DAC_D9	R_DAC_D9	A15
DAC_D3	R_DAC_D3	F13	DAC_D10	R_DAC_D10	B15
DAC_D4	R_DAC_D4	F15	DAC_D11	R_DAC_D11	D15
DAC_D5	R_DAC_D5	E16	DAC_D12	R_DAC_D12	C16
DAC_D6	R_DAC_D6	E15	DAC_D13	R_DAC_D13	B16
DAC_CLK	R_DAC_CLK	A14	DAC_MODE	R_DAC_MODE	H13

4.4 ADC (MCP3202 — U26)

The MCP3202 is a 12-bit, 2-channel SPI ADC. CH0 accepts analog from the on-board LM35 temperature sensor or screw terminal (J8). CH1 connects to potentiometer RV2 for manual analog input.

Signal	FPGA Pin	Description
SPIIm_SCK	G11	ADC clock (shared SPI bus)
SPIIm_MOSI	G12	Data to ADC (DIN)
SPIIm_MISO	G14	Data from ADC (DOUT)
SD_CS (10K pull-up)	H14	ADC chip select (active low)
ADC_CH0	Screw J8 pin 1	External analog input or LM35 output
ADC_CH1	RV2 wiper	On-board pot (0–3.3V range)

4.5 SRAM (IS61WV25616EDBLL — U3)

512K × 16-bit asynchronous SRAM, 3.3V, operating at up to 10 ns access time. All 18 address bits and 16 data bits are connected to FPGA Bank 14.

Signal Group	Nets	FPGA Bank	Bit Width
Address	SRAM_A0 – SRAM_A17	Bank 14	18 bits
Data	SRAM_D0 – SRAM_D15	Bank 14	16 bits
Chip Enable	SRAM_CE	Bank 14	1 bit (active low)
Output Enable	SRAM_OE	Bank 14	1 bit (active low)
Write Enable	SRAM_WE	Bank 14	1 bit (active low)
Upper Byte	SRAM_UB	Bank 14	1 bit (active low)
Lower Byte	SRAM_LB	Bank 14	1 bit (active low)

Nets	FPGA Pin	Nets	FPGA Pin	Nets	FPGA Pin
SRAM_A0	T12	SRAM_A13	T9	SRAM_D8	T10
SRAM_A1	R12	SRAM_A14	T5	SRAM_D9	N9
SRAM_A2	T13	SRAM_A15	R6	SRAM_D10	P9
SRAM_A3	T14	SRAM_A16	R6	SRAM_D11	N6
SRAM_A4	T15	SRAM_A17	R5	SRAM_D12	R10
SRAM_A5	P13	SRAM_D0	R16	SRAM_D13	P10
SRAM_A6	R13	SRAM_D1	P15	SRAM_D14	M12
SRAM_A7	N12	SRAM_D2	P16	SRAM_D15	R11
SRAM_A8	N11	SRAM_D3	N16	SRAM_CE	R15
SRAM_A9	P11	SRAM_D4	M16	SRAM_OE	P6
SRAM_A10	R7	SRAM_D5	M14	SRAM_WE	N13
SRAM_A11	T8	SRAM_D6	N14	SRAM_UB	P8
SRAM_A12	R8	SRAM_D7	P14	SRAM_LB	T7

4.6 Micro SD Card (J6)

SD Signal	Net	FPGA Pin	Description
DAT0	SD_D0	D8 (Bank 15)	MISO / Data bit 0
DAT1	SD_D1	D9 (Bank 15)	Data bit 1
DAT2	SD_D2	B10(Bank 15)	Data bit 2
DAT3/CD	SD_D3	B11 (Bank 15)	CS / Card Detect dual-use
CMD	SD_CMD	B12 (Bank 15)	MOSI / Command

CLK	SD_SCK	C11 (Bank 15)	Clock
DET	SD_DET	C12 (Bank 15)	Card detect switch

4.7 ESP32-C3-MINI-1 (U13)

The ESP32-C3-MINI-1 serves as a Wi-Fi/BLE co-processor. It connects to the FPGA via shared SPI and UART, and has its own USB-C port (J9) for programming.

ESP32 GPIO	Net / Function	FPGA Pin / Connection
GPIO4	SPIIm_MISO	FPGA G14 (Bank 15)
GPIO5	SPIIm_SCK	FPGA G11 (Bank 15)
GPIO6	ESP_CS	FPGA F14 (Bank 15)
GPIO7	SPIIm_MOSI	FPGA G12 (Bank 15)
GPIO20 (U0RXD)	ESP_RX	J13 connector pin 5
GPIO21 (U0TXD)	ESP_TX	J13 connector pin 6
GPIO18 (USB_D-)	ESP_N	J9 USB-C D- (0Ω R20)
GPIO19 (USB_D+)	ESP_P	J9 USB-C D+ (0Ω R21)
GPIO0/1/3/10	ESP_IO0/1/3/10	J13 connector pins 2/3/1/4
EN/CHIP_PU	NRST	RC filter (C90/C91) + SW19 for BOOT

4.8 DC Motor Driver (DRV8833 — U28)

Signal	FPGA Net	FPGA Pin	Description
IN1	IN1	F12 (Bank 14)	Motor A direction input 1
IN2	IN2	H11(Bank 14)	Motor A direction input 2

Current sensing via 200mΩ resistors (R193, R195) at TP10/TP11. Motor M1 connects to AOUT1/AOUT2. Sleep controlled by FPGA (or pulled high for always-on).

4.9 Stepper Motor Driver (L293DD — U16)

Signal	FPGA Net	FPGA Pin (Bank 15)	L293DD Pin
SIG1	SIG1	E12	INPUT_1 (pin 2)
SIG2	SIG2	E13	INPUT_2 (pin 9)
SIG3	SIG3	E11	INPUT_3 (pin 12)
SIG4	SIG4	D11	INPUT_4 (pin 19)

Outputs SM1–SM4 go to J7 screw terminal (5-pin: SM1, SM2, SM3, SM4, +5V). Enable pins are tied high (+5V).

4.10 Relay (K1 — SANYOU SRD Form-C)

Signal	FPGA Net	FPGA Pin	Notes
RLY_IN	RLY_IN	L5 (IO_0_34, Bank 34)	High = relay energized via Q7 (SS8050 NPN)

Relay coil powered from +5V via ferrite FB9. J17 (3-pin screw): NO, COM, NC. Max contacts: 10A @ 125VAC / 10A @ 30VDC.

4.11 Buzzer (BZ1)

Signal	FPGA Net	FPGA Pin	Notes
BZR_OUT	BZR_OUT	K5 (IO_25_35, Bank 35)	Drives BC817 NPN transistor; 100Ω base resistor

5. Configuration & Programming

5.1 Configuration Modes

The configuration mode is selected by three SPDT slide switches on the board:

Mode	SW17 (M2)	SW36 (M1)	SW37 (M0)	Description
Master SPI	0	0	1	Boot from SPI Flash (MX25L3233FM1) — default
JTAG	1	0	1	Load bitstream directly via USB JTAG

5.2 Programming via Vivado

Connect USB-C cable to J1. Use Vivado Hardware Manager to detect the xc7a50t_0 target. For remote lab use, the FT232H enumerates as a JTAG adapter compatible with Xilinx cable drivers.

- Bitstream file: .bit (generated by Vivado implementation)
- Flash programming: Use Vivado's 'Add Configuration Memory Device' → MX25L3233F
- SPI flash address: 0x00000000 (default)

6. XDC Constraints Reference

Below is a complete XDC template covering all major peripherals. Adapt as needed for your Verilog top-level port names.

Clock

```
create_clock -period 41.667 -name sys_clk [get_ports clk_24mhz]
set_property -dict {PACKAGE_PIN D13 IOSTANDARD LVCMOS33} [get_ports clk_24mhz]
```

User LEDs (L1-L8 shown; extend to L16)

```
set_property -dict {PACKAGE_PIN D5 IOSTANDARD LVCMOS33} [get_ports led[0]]
set_property -dict {PACKAGE_PIN A3 IOSTANDARD LVCMOS33} [get_ports led[1]]
set_property -dict {PACKAGE_PIN B4 IOSTANDARD LVCMOS33} [get_ports led[2]]
set_property -dict {PACKAGE_PIN A4 IOSTANDARD LVCMOS33} [get_ports led[3]]
set_property -dict {PACKAGE_PIN E6 IOSTANDARD LVCMOS33} [get_ports led[4]]
set_property -dict {PACKAGE_PIN C13 IOSTANDARD LVCMOS33} [get_ports led[5]]
set_property -dict {PACKAGE_PIN C14 IOSTANDARD LVCMOS33} [get_ports led[6]]
set_property -dict {PACKAGE_PIN D14 IOSTANDARD LVCMOS33} [get_ports led[7]]
```

Slide Switches

```
set_property -dict {PACKAGE_PIN C9 IOSTANDARD LVCMOS33} [get_ports sw[0]]
set_property -dict {PACKAGE_PIN B9 IOSTANDARD LVCMOS33} [get_ports sw[1]]
set_property -dict {PACKAGE_PIN G5 IOSTANDARD LVCMOS33} [get_ports sw[2]]
set_property -dict {PACKAGE_PIN A7 IOSTANDARD LVCMOS33} [get_ports sw[3]]
set_property -dict {PACKAGE_PIN C7 IOSTANDARD LVCMOS33} [get_ports sw[4]]
set_property -dict {PACKAGE_PIN A10 IOSTANDARD LVCMOS33} [get_ports sw[5]]
set_property -dict {PACKAGE_PIN B7 IOSTANDARD LVCMOS33} [get_ports sw[6]]
set_property -dict {PACKAGE_PIN A8 IOSTANDARD LVCMOS33} [get_ports sw[7]]
```

7-Segment Display (MAX7219 SPI)

```
set_property -dict {PACKAGE_PIN J15 IOSTANDARD LVCMOS33} [get_ports seg_din]
set_property -dict {PACKAGE_PIN J16 IOSTANDARD LVCMOS33} [get_ports seg_load]
set_property -dict {PACKAGE_PIN H12 IOSTANDARD LVCMOS33} [get_ports seg_clk]
```

LCD (16x2)

```
set_property -dict {PACKAGE_PIN G4 IOSTANDARD LVCMOS33} [get_ports lcd_rs]
set_property -dict {PACKAGE_PIN H3 IOSTANDARD LVCMOS33} [get_ports lcd_rw]
set_property -dict {PACKAGE_PIN E1 IOSTANDARD LVCMOS33} [get_ports lcd_en]
set_property -dict {PACKAGE_PIN G2 IOSTANDARD LVCMOS33} [get_ports lcd_d[0]]
set_property -dict {PACKAGE_PIN G1 IOSTANDARD LVCMOS33} [get_ports lcd_d[1]]
set_property -dict {PACKAGE_PIN H5 IOSTANDARD LVCMOS33} [get_ports lcd_d[2]]
set_property -dict {PACKAGE_PIN H4 IOSTANDARD LVCMOS33} [get_ports lcd_d[3]]
set_property -dict {PACKAGE_PIN J5 IOSTANDARD LVCMOS33} [get_ports lcd_d[4]]
set_property -dict {PACKAGE_PIN J4 IOSTANDARD LVCMOS33} [get_ports lcd_d[5]]
set_property -dict {PACKAGE_PIN H2 IOSTANDARD LVCMOS33} [get_ports lcd_d[6]]
set_property -dict {PACKAGE_PIN H1 IOSTANDARD LVCMOS33} [get_ports lcd_d[7]]
```

Buzzer

```
set_property -dict {PACKAGE_PIN K5 IOSTANDARD LVCMOS33} [get_ports buzzer]
```

Relay

```
set_property -dict {PACKAGE_PIN L5 IOSTANDARD LVCMOS33} [get_ports relay_in]
```

SPI (Shared Bus)

```
set_property -dict {PACKAGE_PIN G11 IOSTANDARD LVCMOS33} [get_ports spi_sck]
set_property -dict {PACKAGE_PIN G14 IOSTANDARD LVCMOS33} [get_ports spi_mosi]
set_property -dict {PACKAGE_PIN G12 IOSTANDARD LVCMOS33} [get_ports spi_miso]
set_property -dict {PACKAGE_PIN H14 IOSTANDARD LVCMOS33} [get_ports sd_cs]
```

```
## PMOD (IO_0-IO_7 shown)
set_property -dict {PACKAGE_PIN T2 IOSTANDARD LVCMOS33} [get_ports pmod[0]]
set_property -dict {PACKAGE_PIN R3 IOSTANDARD LVCMOS33} [get_ports pmod[1]]
set_property -dict {PACKAGE_PIN T3 IOSTANDARD LVCMOS33} [get_ports pmod[2]]
set_property -dict {PACKAGE_PIN T4 IOSTANDARD LVCMOS33} [get_ports pmod[3]]
set_property -dict {PACKAGE_PIN M1 IOSTANDARD LVCMOS33} [get_ports pmod[4]]
set_property -dict {PACKAGE_PIN M2 IOSTANDARD LVCMOS33} [get_ports pmod[5]]
set_property -dict {PACKAGE_PIN P1 IOSTANDARD LVCMOS33} [get_ports pmod[6]]
set_property -dict {PACKAGE_PIN R1 IOSTANDARD LVCMOS33} [get_ports pmod[7]]
```

7. Quick-Start Guide

7.1 Hardware Setup

1. Connect 15V DC power supply to J2 (barrel jack). Verify the PWR 5V0, PWR 3V3, PWR 1V8, and PWR 1V0 LEDs all illuminate.
2. Set configuration mode switches: SW17 (M2) = OFF, SW36 (M1) = OFF, SW37 (M0) = ON for Master SPI mode.
3. Connect USB-C cable to J1 (FT232H). Install Xilinx FTDI drivers if not present.
4. Open Vivado → Hardware Manager → Open Target → Auto Connect. The xc7a50t_0 device should appear.
5. Program Device → select your .bit file → Program.
6. DONE LED (D2) should illuminate after successful configuration.

7.2 Minimal Blink XDC + Verilog

Use clock pin D13 (24 MHz), LEDs B7–B5, switches A7–A8. A 24-bit prescaler gives ~1.4 Hz blink at 24 MHz.

7.3 Re-programming the Board

- Press BTN2 (PROGRAM) to pulse PROGRAM_B low — this triggers reconfiguration from SPI flash.
- For a soft reset, toggle PROGRAM_B via Vivado TCL: `set_property PROGRAM.RESET_AFTER_PROGRAM true`.